

SimOpt Solver Competition

September 9, 2026

The 2026 SimOpt Solver Competition <https://github.com/simopt-admin/simopt-competition-26> entails designing and building a solver suitable for inclusion in the SimOpt codebase for tackling optimization problems with integer-ordered decision variables, where the objective function is (approximately) evaluated through a stochastic simulation. A special case of this class of problems is the [tele-operator problem](#), which arises in the realm of autonomous vehicles (AVs). The full details of the tele-operator problem are provided separately in a Markdown file, as is an implementation of the tele-operator problem that works with SimOpt code. Entries are judged on how well and how quickly they solve some (held out) instances of the tele-operator problem and some other (held out) problems with similar structure, along with the quality of the documentation of the solver.

The general form of the problems you are tackling is as follows. The goal is to minimize a function $f(x) = \mathbb{E}f(x, Y)$, where x is a vector of integer-ordered decision variables, Y is a random element, $f(x, Y)$ represents the (real-valued) simulation logic that combines x and Y to yield a single output, and $\mathbb{E}$ denotes expectation. Here, x is [box constrained](#). Box constraints limit each decision variable to a finite value within an interval $[a, b]$, where a or b can be infinite. For example, a variable taking nonnegative integer values would have $a = 0$ and $b = \infty$. These problems are otherwise unconstrained. The (exact) function $f(\cdot)$ cannot be computed exactly, so is approximated through Monte Carlo simulation by

$$f(x) \approx \frac{1}{n} \sum_{i=1}^n f(x, Y_i),$$

where $Y_1, Y_2, \dots, Y_n$ are i.i.d. realizations of the random element Y .

An entry is submitted by a team of authors and consists of

1. Python code for your solver that works with the SimOpt interface — see <http://simopt.org>;
2. An `.md/.rst` file that describes your solver, similar to those available for other solvers already found [on SimOpt](#);
3. Clearance (i.e., the team's permission) for SimOpt to add your solver to the SimOpt library of solvers; and
4. Names and contact information for all team members.

Entries are due on November 7, 2026. Submit your entry as a zip file by email to sgh9@cornell.edu with the subject “SimOpt 2026 Competition”.

Teams can be of any size. A person cannot be a member of multiple teams. Teams can submit only one solver.

Entries will be run against one another in a SimOpt simulation experiment consisting of all running entered solvers and several (held out) test problems, coded using the SimOpt interface. Some of the test problems will be of the tele-operator form with customized input parameters. We will estimate the mean μ of the mean area under normalized progress curves when solvers are initiated from a fixed starting solution as computed by SimOpt, averaged across all test problems, equally weighted.

The winning entry(ies) will be the one(s) that minimizes the estimated value of μ over the held-out test problems. The winning entry(ies) will be announced at the 2026 Winter Simulation Conference in Glasgow. The winning entry will receive \$600 and the second-place entry will receive \$400. In the event of (statistical) ties for first place, ties will be broken by the quality of the `.rst` file that describes the entry. If ties for first place persist, then the prize money of \$1000 will be shared equally between winning entries. In the event of a unique first-place team and ties for second place, the second-place prize money of \$400 will be shared equally between second-place entries.

Queries may be sent to Shane Henderson (sgh9@cornell.edu).